

Project No. : 24167

Borehole Number : OW-32

Page 1 of 2

Project : MRT Orange Line (West Section)

Borehole Location : Tunnel (Sta. 15+450 WB)

Co-ordinate (N) : 1,521,296

Ground Elevation (m) : +101.160 m.MSL

Co-ordinate (E) : 663,934

G.W.L. (m) : 1.0 (below ground level)

Date Started : 15 Oct 2024

Total Depth (m) : 60.45

Date Finished : 21 Oct 2024

Depth (m)	SAMPLE			SUBSURFACE PROFILE		Atterberg Limits (%) 20 40 60 80	Unit Weight (kN/m³) 5 15	Suc (kN/m²) Δ						
	Symbol	Type	Number	Description	Symbol			SPT-N □ (Blows/300mm) □						
								20	40	60	80			
0				Ground Surface										
1				Fill Material / Top Soil	0.00									
2	■	ST	1											
3	■	ST	2											
4	■	ST	3	Medium stiff CLAY, grey, high plasticity (CH)	3.50							26.5		
5	■	ST	4	Very soft to soft CLAY, trace of sand, grey, hig plasticity (CH)	5.50							12.8		
6	■	ST	5										16.5	
7	■	ST	6										5.7	
8	■	ST	7										18.6	
9	■	ST	8										17.9	
10	■	ST	9	Medium stiff CLAY, grey, high plasticity (CH)	13.00							40.4		
11	■	ST	10	Very stiff silty CLAY, trace to few sand, grey and greyish brown, high to low plasticity (CH, CL)	14.50								112.2	
12	■	SS	1										26	
13	■	SS	2										19	
14	■	SS	3									25		
15	■	SS	4	Hard silty CLAY, trace of sand, greyish brown, low plasticity (CL)	19.00							32		
16	■	SS	5	Stiff sandy CLAY, greyish brown, low plasticity (CL)	20.50							15		
17	■	SS	6	Very stiff silty CLAY, trace of sand, greyish brown, low plasticity (CL)	22.00							24		
18	■	SS	7										45	
19	■	SS	8	Hard silty CLAY, few sand, greyish brown, low plasticity (CL)	23.50							36		
20	■	SS	9	Very stiff silty CLAY, trace of sand, greyish brown, low plasticity (CL)	26.50							21		
21	■	SS	10	Hard silty CLAY, few sand, brown, low plasticity (CL)	28.00							42		
22	■	SS	11											78/275
23	■	SS	12	Medium dense to very dense silty SAND / SAND with silt, greyish brown to brown, fine to medium grained (SM, SP-SM)	29.50							80/250		
24	■	SS	13										28	
25	■	SS	14										80/275	
26	■	SS	15										67/200	
27	■	SS	16										50/125	
28	■	SS	17										78/275	
29	■	SS												

Project : MRT Orange Line (West Section)

Depth (m)	SAMPLE			SUBSURFACE PROFILE		Atterberg Limits (%) 20 40 60 80	Unit Weight (kN/m³) 5 15	Suc (kN/m²) Δ 20 40 60 80 Δ			
	Symbol	Type	Number	Description	Symbol			SPT-N □ (Blows/300mm) □ 20 40 60 80			
40	SS	18	Hard sandy CLAY, grey, low plasticity (CL)	40.00	□	●	50/100				
41	SS	19	Very dense silty SAND, greyish brown, fine grained (SM)	41.50	□	●	50/125				
42	SS	20			□	●	50/100				
43	SS	21	Hard silty CLAY, little sand for SS-23, greyish brown, low plasticity (CL)	44.50	□	●	50/150				
44	SS	22			□	●	50/100				
45	SS	23			□	●	50/100				
46	SS	24	Very dense silty SAND, greyish brown, fine grained (SM)	49.00	□	●	50/125				
47	SS	25	Hard sandy CLAY, greyish brown, low plasticity (CL)	50.50	□	●	50/100				
48	SS	26	Very dense silty SAND, greyish brown, fine grained (SM)	52.00	□	●	50/100				
49	SS	27			□	●	50/125				
50	SS	28			□	●	50/75				
51	SS	29	Very dense SAND with silt, grey, medium grained (SP-SM)	56.50	□	●	50/125				
52	SS	30			□	●	50/75				
53	SS	31			□	●	50/75				
54			End of Borehole	60.45							
55											
56											
57											
58											
59											
60											
61											
62											
63											
64											
65											
66											
67											
68											
69											
70											
71											
72											
73											
74											
75											
76											
77											
78											
79											
80											
81											
82											
83											

Thin wall tube

Split spoon

Rock core

Wash

Auger

Atterberg Limits : PL Wn LL

Suc : Undrained shear strength from UC test

SPT-N : Field SPT-N (donut hammer with pulley and use split spoon sampling with liner)